

Investigative Data Pipeline Design

A scalable architecture for reconciling CSV and device-log evidence, preserving integrity, and supporting investigator queries.

1. Core Idea

The proposed system ingests evidence from CSV files and device logs, preserves the original evidence, verifies integrity with cryptographic hashes, normalizes and reconciles the data in Python, stores structured records in MySQL, and uses Gemma to translate natural-language investigator questions into safe read-only SQL queries.

2. Recommended Architecture

DATA SOURCES	CSV files GPS logs Wi-Fi logs device logs
RAW EVIDENCE STORE	Original files preserved exactly as received
INTEGRITY + PROVENANCE	SHA-256 hash, source, received time, import job
PYTHON ETL	Parse, validate, normalize timestamps, identifiers and locations
ENTITY RESOLUTION	Map person <-> device <-> account <-> badge
MYSQL	Events, persons, devices, sources, evidence, hashes
QUERY SAFETY LAYER	SELECT-only validation + read-only DB account
GEMMA	Natural language -> SQL
INVESTIGATOR OUTPUT	Answer + raw evidence + source + confidence

3. Preserve the Original Evidence

Do not overwrite or replace the source files. Keep every original CSV or log exactly as received, then create transformed records separately. This lets investigators trace any normalized result back to the original evidence.

```
Original record:
2026/10/04 02:13:55 EST, Device-AA31, 43.71, -79.40

Normalized record:
timestamp_utc: 2026-10-04T06:13:55Z
device_id: AA31
latitude: 43.71
longitude: -79.40
```

4. Evidence Integrity and Hashing

Generate a SHA-256 hash for each original file when it enters the system. Later, hashing the file again and comparing the result can show whether the file changed. A hash by itself does not prove who supplied the file, so store provenance metadata alongside it.

Field	Purpose
evidence_id	Unique evidence record
original_filename	Original file name
source_system	Where the evidence came from
received_at	When the evidence entered the system
received_by	Who or what process received it
sha256	Integrity fingerprint
original_path	Location of preserved raw evidence
import_job_id	Links evidence to the processing run

5. Normalize More Than Just Timestamp Format

For timestamps, preserve the original timestamp and timezone while also creating a normalized UTC value. The transformation layer should also standardize device identifiers, location formats, missing values, and source-specific field names.

6. Entity Resolution

This is one of the most important missing pieces. Device logs usually tell you where a device, account, badge, or vehicle was - not automatically where a person was. The system needs a controlled mapping between people and the identifiers associated with them.

```

Person: John Smith
Phone -> Device 8F22
Wi-Fi -> MAC A1:B2:C3
Badge -> 4421
Vehicle -> V72

```

7. Model the Data as Events

A common event structure makes records from different sources easier to reconcile and query.

Field	Meaning
event_id	Unique event
person_id	Resolved person, if known
device_id	Associated device
timestamp_utc	Normalized time
latitude / longitude	Location coordinates, when available
location_name	Human-readable location
event_type	GPS, Wi-Fi, badge, etc.
source_id	Original evidence source
accuracy_meters	Estimated spatial accuracy
confidence_level	Reliability / certainty
original_record_id	Traceback to raw evidence

8. Represent Location Confidence

Not every location source has the same precision. The system should not present approximate evidence as an exact location.

Source	Typical interpretation
GPS	High precision, depending on device conditions
Cell tower	Regional / lower precision
Wi-Fi association	Often building or network-level
Security badge	Door or access-point level
IP address	Usually poor evidence of precise physical location

9. Gemma Query Layer

Gemma can translate investigator questions into SQL, but it should not receive unrestricted database permissions. Use a read-only database account and validate generated SQL before execution. Ideally, expose a small set of investigator-focused views instead of the entire schema.

```
Investigator:
"Where was Person A between 8 PM and 10 PM?"

Gemma -> SQL:
SELECT timestamp_utc, latitude, longitude, source_id
FROM investigator_events_view
WHERE person_id = 41
AND timestamp_utc BETWEEN ...
```

```
ORDER BY timestamp_utc;
```

Reject: DELETE, UPDATE, INSERT, DROP, ALTER and any other write or administrative statements.

10. Audit Trail

Record who queried the system, what natural-language question they asked, the SQL that was generated, when the query ran, and which records were returned. This gives the system accountability and makes investigative use easier to review.

11. Investigator-Facing Output

Do not return only a model-written conclusion. Present a concise answer together with the underlying records, source evidence, timestamps, and confidence information so the investigator can independently verify the result.

Possible match:

Device assigned to John Smith was recorded near 100 Main Street between 21:03 and 21:17.

Supporting evidence:

- GPS log - 21:03:11
- Wi-Fi connection - 21:08:43
- Badge entry - 21:12:02

Confidence: High

[\[View raw records\]](#)

12. Final Recommended Flow

CSV / device logs -> immutable raw evidence -> SHA-256 + provenance -> Python ETL -> entity resolution -> MySQL -> read-only query API -> Gemma natural-language-to-SQL -> validated SQL -> investigator result with raw evidence, source and confidence.